EE 616 Faculty Review Packet

Vision-Based Leader-Follower Coordination for Warehouse Material Transport

Project identification

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Faculty reviewer: Professor Masudul Imtiaz
Course: EE 616-57, section 9189, four credits
Current stage: Completed simulation package awaiting faculty review

Project summary

This project implements and evaluates a warehouse convoy in which each independently driven follower observes the robot immediately ahead. A follower uses a forward RGB camera and local wheel odometry. YOLOv8n target detection, calibrated monocular geometry, a relative EKF, bounded formation control, and explicit visual-loss states produce the follower command.

The leader owns a repeatable test route. Follower 1 tracks the leader, Follower 2 tracks Follower 1, and each later follower tracks its immediate predecessor. Gazebo position truth is restricted to evaluation and never enters perception, estimation, supervision, or control.

Current claim boundary

The accepted evidence covers ROS 2 and Gazebo simulation on one constrained laptop. It does not demonstrate physical robots, real-camera generalization, person safety, industrial certification, or production readiness.

Review path

- Project definition and architecture: pages 2-3.
- Accepted results and timing: pages 4-5.
- Demonstrations, screenshots, code map, limitations, and requested guidance: pages 6-8.

1. Problem and Project Scope

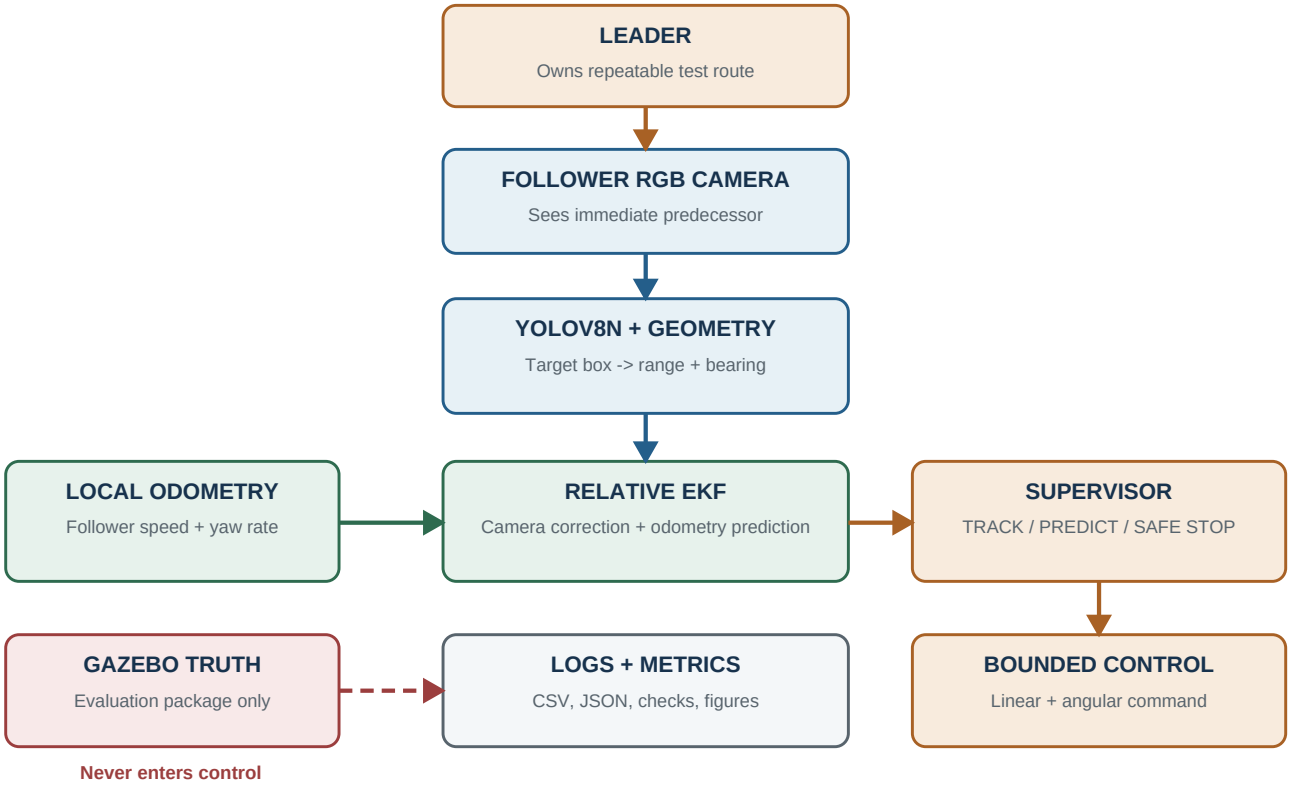
Research question

How accurately and reliably can an independently driven follower maintain a requested distance from the robot directly ahead using a forward RGB camera and local wheel odometry?

Why predecessor following

- Each follower uses only local sensing and its own controller instance.
- The leader route does not need to be transmitted to every follower.
- The same follower pipeline can be repeated under isolated ROS 2 namespaces.
- The main tradeoff is that local errors and disturbances can propagate toward the rear.

Implemented architecture



The known rear target is 0.70 m wide and 0.90 m high. The fixed target size resolves monocular scale. This is a controlled marker-like design and not general robot recognition.

Leader boundary

The validated leader follows deterministic straight, turn, sharp-turn, stop, and combined profiles. Nav2, SLAM, A*, RRT-Connect, and autonomous task assignment are not part of the accepted stack.

2. Perception, Estimation, and Control

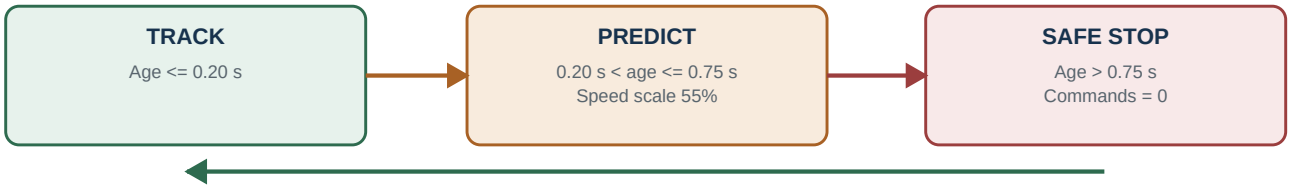
Camera measurement

YOLOv8n selects the highest-confidence predecessor-target box. Horizontal box position determines bearing. Bounding-box height, focal length, and known target height determine forward depth. A calibration scale and offset correct the final planar range.

Relative EKF

The EKF state contains range, bearing, range rate, and bearing rate. Camera measurements correct the state. The follower's own forward speed and yaw rate support prediction between measurements. Prediction is limited in time because uncertainty grows during visual loss.

Supervisor



Main parameters

Parameter	Value	Meaning
Requested range	1.50 m	Camera-to-predecessor target distance
RGB camera	640 x 480; 70 deg HFOV	Forward follower image
Camera rate	15 Hz configured	Sensor update setting
Controller	20 Hz / 50 ms	Follower command update
Linear bound	0.00 to 0.60 m/s	No reverse command in the accepted controller
Angular bound	+/- 0.80 rad/s	Bounded steering command
PREDICT scale	55%	Reduced forward speed during short visual loss

Evaluation-only truth

Gazebo pose truth is read only by the evaluation package to calculate errors and checks. Dependency tests and topic boundaries prevent pose truth from becoming an operational controller input.

3. Validation Sequence and Behavior Results

Gate	Accepted result	Scope boundary
Environment	ROS 2 Jazzy, Gazebo Sim 8.15.0, namespaced cameras, and NVIDIA access verified.	Canonical container on one target laptop.
YOLOv8n v3 measurement	1,620/1,620 valid fully visible samples; 0.0323 m range RMSE; 0.1569 deg bearing MAE; 9.02 ms inference p95.	Two controlled sealed scenes and fixed-size target.
One pair	Six runs passed. Worst RMSE: 0.0558 m straight and 0.1005 m gradual turn. Minimum TRACK: 99.1%.	Nominal simulated pair.
Incremental chain	Twelve staged runs passed. Three-follower worst F1/F2/F3 RMSE: 0.1004/0.0739/0.0523 m.	Nominal matrix; rearward amplification not observed.
Controlled disturbance	Eighteen pair-first runs passed. Combined worst F1/F2/F3 RMSE: 0.1007/0.1242/0.1403 m.	Synthetic blackout and bounded yaw-command bias; rearward growth observed in this matrix.
Timing	Three combined-chain timing runs passed with minimum real-time factor 0.9996.	One-host simulation timing.

Correct interpretation

- The pair is the minimum validated scientific unit.
- The nominal chain did not show rearward amplification.
- The combined disturbance matrix did show rearward error growth.
- Neither matrix proves general string stability or instability.
- All accepted behavior matrices recorded zero collision samples under the evaluator threshold.

Zero collision samples is not equivalent to continuous collision avoidance, person safety, industrial certification, or a production-ready system.

Retained failed evidence

The first two YOLOv8n checkpoints remain recorded failures. The v3 stage corrected an RGB/BGR runtime mismatch and stale-frame dataset labels. The failed checkpoints were not hidden or used for control.

4. Metrics, Timing, and Laptop Resources

Metric definitions

Metric	Definition	Interpretation
Spacing error	True range - 1.50 m	Positive is too far; negative is too close.
RMSE	Square root of mean squared error	Typical magnitude with stronger penalty for large deviations.
Bearing MAE	Mean absolute angular error	Average horizontal direction error.
TRACK fraction	TRACK samples / evaluated samples	Fraction of evaluated behavior using a current measurement.
Reacquisition	Event end to return to TRACK	Recovery of normal visual tracking.
p95	95th percentile	95% of values are at or below it; rare larger values may remain.
Real-time factor	Simulated time / wall time	Near 1.0 means the simulation keeps pace with real time.
Collision sample	Center distance below 0.65 m	A sampled evaluator rule, not safety certification.

Worst accepted p95 timing values

Timing path	Value
Camera message to perception measurement	23.09 ms
Valid measurement to controller command	48.35 ms
Controller to post-proxy command	2.89 ms
YOLOv8n inference	20.34 ms
Control period	50.0 ms

Peak timing-gate resource observations

Resource	Peak
Container memory	3,584 MiB
Visible GPU memory	642 MiB of 4,096 MiB
GPU utilization	60%
CPU	443.1% of one-core equivalent across 12 logical CPUs

These values support feasibility for the declared one-laptop simulation. They exclude real camera drivers, networks, motors, wheel slip, and physical actuator delay.

5. Demonstrations and Reproduction

Recorded presentation runs

Video	What it shows	Recorded summary
Pair turn	One leader and one follower during a gradual turn.	0.104 m spacing RMSE; 97.3% TRACK; 50 ms control p95; zero collision samples.
Three-follower chain	One leader and three independently driven followers.	0.058/0.051/0.050 m RMSE; 95%/99%/97% TRACK; zero collision samples.
Recovery pair	1.3 s full-frame RGB blackout and state recovery.	PREDICT and SAFE STOP observed; zero stop command; 0.40 s reacquisition; 1.00 s spacing recovery.

Live commands

```
make professor-demo PROFESSOR_DEMO=pair_turn
make professor-demo PROFESSOR_DEMO=chain_straight
make professor-demo PROFESSOR_DEMO=recovery
```

Evidence reproduction commands

```
make environment-check-gpu
make pair-gate
make chain-gate
make disturbance-gate
make timing-gate
python3 work/gate7_package/verify_frozen_evidence.py
```

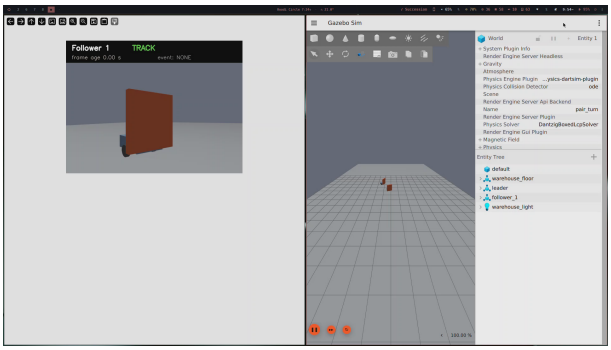
The experiment commands intentionally overwrite their corresponding working result directories. The frozen Gate 7 package should be preserved before intentionally collecting a new experimental revision.

Evidence hierarchy

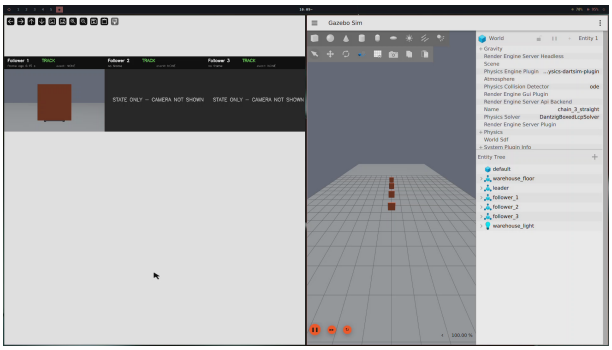
- Machine-readable frozen summaries and raw rows are the primary numerical evidence.
- The three presentation videos show behavior but do not create new scientific gates.
- The earlier toy simulator remains preliminary design evidence and is not mixed with Gazebo results.

6. Presentation Screenshots

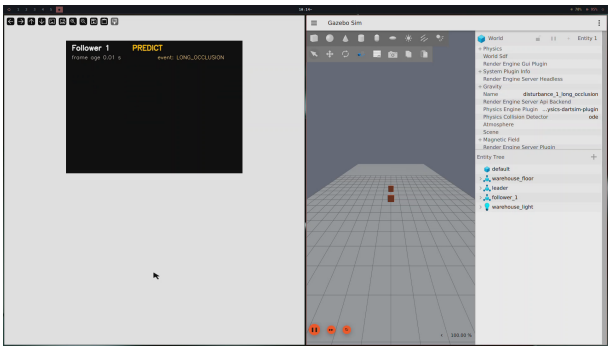
These full-resolution frames were extracted from the three recorded presentation videos. They are visual aids, not separate experimental repetitions.



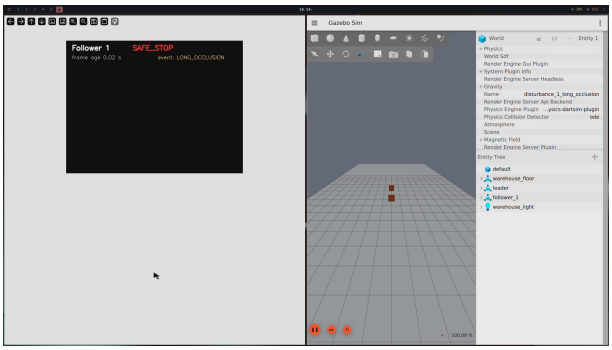
Pair gradual turn: Follower 1 in TRACK.



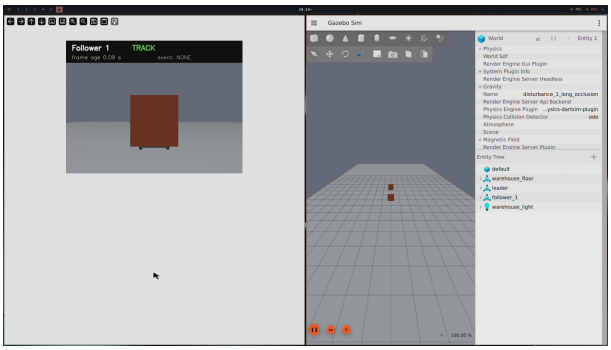
Three-follower chain: all follower supervisors in TRACK.



Synthetic RGB blackout: PREDICT.



Continued synthetic blackout: SAFE STOP.



Target returned: TRACK reacquired.

The black image is created by the controlled disturbance proxy. Gazebo continues to simulate robot motion. When the target image returns, the supervisor reacquires TRACK.

Recovery sequence interpretation.

The read-only dashboard displays one representative RGB stream and every follower state. It does not publish control commands.

7. Code Map, Limitations, and Requested Guidance

Primary code locations

Responsibility	Location
Robot and camera construction	`work/ros2_ws/src/ee616_simulation/ee616_simulation/pair_world.py`
YOLO and range/bearing	`work/ros2_ws/src/ee616_perception/ee616_perception/yolo_target.py`
EKF and supervisor	`work/ros2_ws/src/ee616_control/ee616_control/control_core.py`
Follower ROS node	`work/ros2_ws/src/ee616_control/ee616_control/follower_node.py`
Leader test route	`work/ros2_ws/src/ee616_control/ee616_control/leader_node.py`
Evaluation truth boundary	`work/ros2_ws/src/ee616_evaluation/ee616_evaluation/pair_gate_evaluator.py`
Presentation launch	`work/ros2_ws/src/ee616_evaluation/launch/professor_demo.launch.py`

Not yet verified

- Physical robots, real camera calibration, and uncontrolled lighting.
- General predecessor appearance without the known fixed-size target.
- Wheel slip, physical motor faults, and real network timing.
- Safety-rated obstacle avoidance, people, and industrial certification.
- Autonomous leader localization, global planning, and task assignment.
- General string stability.

Requested faculty guidance

The simulation-stage implementation is complete. The next approved stage could deepen perception realism, add a known-map Nav2 leader baseline, strengthen control and string-stability analysis, or define a small physical pair experiment with a separate safety and hardware scope.

Related leader-follower and camera-based systems already exist. The contribution claimed here is the implementation and staged evaluation of this specific constrained warehouse architecture, not the invention of leader-follower control.